

INTERVIEW THEORY QUESTION BANK

300+ Frequently Asked Questions with Short, Interview-Friendly Answers

Subjects Covered: OOPs • DBMS • SQL • Operating Systems • Computer Networks • DSA • Software Engineering • Computer Organization • Python/Java • Web Technologies

Total Questions: 350

Use this as a rapid revision guide. For interviews, first learn the definition, then prepare one example and one follow-up point for important topics.

CODE WITH SWAROOP

1. OOPs

35 interview questions

Q1. What is OOP?

Answer: Object-Oriented Programming is a programming paradigm that organizes software around objects containing data and behavior.

Q2. What is a class?

Answer: A class is a blueprint that defines the properties and methods of objects.

Q3. What is an object?

Answer: An object is an instance of a class with its own state and behavior.

Q4. What is encapsulation?

Answer: Encapsulation bundles data and methods together and restricts direct access to internal data.

Q5. What is abstraction?

Answer: Abstraction hides implementation details and exposes only the essential functionality.

Q6. What is inheritance?

Answer: Inheritance allows a class to acquire properties and methods of another class.

Q7. What is polymorphism?

Answer: Polymorphism allows the same interface or method name to behave differently for different objects.

Q8. What is method overloading?

Answer: Defining multiple methods with the same name but different parameter lists is overloading.

Q9. What is method overriding?

Answer: A subclass provides its own implementation of a method inherited from its parent class.

Q10. Overloading vs overriding?

Answer: Overloading is usually compile-time polymorphism; overriding is runtime polymorphism.

Q11. What is a constructor?

Answer: A constructor initializes an object when it is created.

Q12. What is a destructor?

Answer: A destructor is used for cleanup when an object is destroyed; exact behavior depends on the language.

Q13. What is an interface?

Answer: An interface defines a contract of methods that implementing classes must provide.

Q14. Abstract class vs interface?

Answer: An abstract class can contain state and implemented methods; an interface primarily defines a contract, with language-specific exceptions.

Q15. What is multiple inheritance?

Answer: Multiple inheritance means a class inherits from more than one parent class.

Q16. What is composition?

Answer: Composition models a strong has-a relationship where one object owns or contains another.

Q17. Aggregation vs composition?

Answer: Aggregation is a weaker has-a relationship; composition implies stronger ownership and lifecycle dependency.

Q18. What is association?

Answer: Association is a general relationship between two classes or objects.

Q19. What is dynamic binding?

Answer: Dynamic binding resolves an overridden method at runtime based on the actual object.

Q20. What is static binding?

Answer: Static binding resolves a method call at compile time.

Q21. What is coupling?

Answer: Coupling measures how strongly one module depends on another.

Q22. What is cohesion?

Answer: Cohesion measures how closely related the responsibilities inside a module are.

Q23. What is SOLID?

Answer: SOLID is a set of five principles for designing maintainable object-oriented software.

Q24. What is SRP?

Answer: Single Responsibility Principle says a class should have one reason to change.

Q25. What is the Open/Closed Principle?

Answer: Software entities should be open for extension but closed for modification.

Q26. What is Liskov Substitution Principle?

Answer: Subtypes should be usable wherever their base types are expected without breaking correctness.

Q27. What is Dependency Inversion?

Answer: High-level modules should depend on abstractions rather than concrete implementations.

Q28. What is a virtual function?

Answer: A virtual function supports runtime dispatch so an overridden implementation can be selected dynamically.

Q29. What is message passing?

Answer: Objects communicate by invoking methods or sending messages to one another.

Q30. Why use OOP?

Answer: OOP improves modularity, reuse, maintainability, and modeling of complex systems.

Q31. What is a default constructor?

Answer: A constructor supplied automatically by a language/compiler under certain conditions when no constructor is explicitly defined.

Q32. What is an immutable object?

Answer: An object whose state cannot be changed after creation.

Q33. What is an is-a relationship?

Answer: An is-a relationship usually represents inheritance, where a subtype is a kind of its parent type.

Q34. What is a has-a relationship?

Answer: A has-a relationship represents composition or aggregation between objects.

Q35. What is dependency injection?

Answer: Dependency injection supplies an object's dependencies from outside instead of constructing them internally.

2. DBMS

35 interview questions

Q1. What is a DBMS?

Answer: A DBMS is software used to create, store, organize, retrieve, and manage data in databases.

Q2. DBMS vs RDBMS?

Answer: RDBMS stores data in related tables and typically follows relational principles; DBMS is the broader term.

Q3. What is a database?

Answer: A database is an organized collection of data that can be efficiently accessed and managed.

Q4. What is a table?

Answer: A table stores related data in rows and columns.

Q5. What is a primary key?

Answer: A primary key uniquely identifies each row and cannot contain NULL values.

Q6. What is a foreign key?

Answer: A foreign key references a key in another table to maintain referential integrity.

Q7. What is a candidate key?

Answer: A candidate key is a minimal attribute set capable of uniquely identifying a row.

Q8. What is a super key?

Answer: A super key is any attribute set that uniquely identifies a row, possibly with extra attributes.

Q9. What is normalization?

Answer: Normalization organizes data to reduce redundancy and update anomalies.

Q10. What is 1NF?

Answer: First Normal Form requires atomic values and no repeating groups.

Q11. What is 2NF?

Answer: Second Normal Form is 1NF plus removal of partial dependency on a composite key.

Q12. What is 3NF?

Answer: Third Normal Form is 2NF plus removal of transitive dependency.

Q13. What is BCNF?

Answer: BCNF requires every determinant to be a candidate key.

Q14. What is denormalization?

Answer: Denormalization intentionally adds redundancy to improve read performance in selected cases.

Q15. What is a transaction?

Answer: A transaction is a logical unit of database work that should be completed reliably.

Q16. What are ACID properties?

Answer: Atomicity, Consistency, Isolation, and Durability ensure reliable transactions.

Q17. What is atomicity?

Answer: Atomicity means a transaction happens completely or not at all.

Q18. What is consistency?

Answer: Consistency means a transaction moves the database from one valid state to another.

Q19. What is isolation?

Answer: Isolation controls how concurrent transactions interact with each other.

Q20. What is durability?

Answer: Durability means committed changes survive failures.

Q21. What is a view?

Answer: A view is a virtual table based on a query.

Q22. What is an index?

Answer: An index is a data structure that speeds up searches but consumes storage and can slow writes.

Q23. Clustered vs non-clustered index?

Answer: A clustered index determines the physical/logical row ordering in systems that support it; a non-clustered index is a separate lookup structure.

Q24. What is a stored procedure?

Answer: A stored procedure is a named collection of database statements stored and executed by the DBMS.

Q25. What is a trigger?

Answer: A trigger automatically executes in response to specified database events.

Q26. What is a deadlock?

Answer: A deadlock occurs when transactions wait indefinitely for resources held by each other.

Q27. What is referential integrity?

Answer: It ensures foreign-key references point to valid related rows or follow defined NULL rules.

Q28. What is a schema?

Answer: A schema describes the logical structure of database objects and their relationships.

Q29. What is data independence?

Answer: Data independence means changing one schema level without requiring changes at higher levels.

Q30. What is an ER diagram?

Answer: An Entity-Relationship diagram visually models entities, attributes, and relationships.

Q31. What is cardinality?

Answer: Cardinality describes the number of rows or relationship instances, depending on context.

Q32. What is a composite key?

Answer: A composite key uses multiple columns together to uniquely identify a row.

Q33. What is a database transaction log?

Answer: A transaction log records changes needed for recovery and durability.

Q34. What is concurrency control?

Answer: Concurrency control coordinates simultaneous transactions to preserve correctness.

Q35. What is a checkpoint?

Answer: A checkpoint records a recovery point so the DBMS can reduce recovery work after failure.

3. SQL

35 interview questions

Q1. What is SQL?

Answer: SQL is the language used to define, query, manipulate, and control data in relational databases.

Q2. What is DDL?

Answer: DDL defines database structures; common commands include CREATE, ALTER, DROP, and TRUNCATE.

Q3. What is DML?

Answer: DML modifies data; common commands include INSERT, UPDATE, and DELETE.

Q4. What is DQL?

Answer: DQL is commonly used to describe SELECT queries that retrieve data.

Q5. What is DCL?

Answer: DCL controls permissions using commands such as GRANT and REVOKE.

Q6. What is TCL?

Answer: TCL manages transactions using commands such as COMMIT and ROLLBACK.

Q7. DELETE vs TRUNCATE?

Answer: DELETE removes selected rows and can use a WHERE clause; TRUNCATE removes all rows more directly and is DBMS-specific in details.

Q8. DROP vs TRUNCATE?

Answer: DROP removes the object itself; TRUNCATE removes rows while retaining the table structure.

Q9. WHERE vs HAVING?

Answer: WHERE filters rows before grouping; HAVING filters groups after aggregation.

Q10. What is a JOIN?

Answer: A JOIN combines rows from related tables based on a matching condition.

Q11. INNER JOIN?

Answer: Returns rows having matches in both joined tables.

Q12. LEFT JOIN?

Answer: Returns all rows from the left table and matching rows from the right table.

Q13. RIGHT JOIN?

Answer: Returns all rows from the right table and matching rows from the left table.

Q14. FULL OUTER JOIN?

Answer: Returns matched rows plus unmatched rows from both sides where supported.

Q15. What is a self join?

Answer: A self join joins a table with itself using aliases.

Q16. What is a subquery?

Answer: A subquery is a query nested inside another SQL statement.

Q17. What is a correlated subquery?

Answer: A correlated subquery references columns from the outer query and is evaluated in relation to outer rows.

Q18. What is GROUP BY?

Answer: GROUP BY groups rows so aggregate functions can calculate values per group.

Q19. What is ORDER BY?

Answer: ORDER BY sorts query results by one or more expressions.

Q20. What is DISTINCT?

Answer: DISTINCT removes duplicate result rows.

Q21. What is an aggregate function?

Answer: Functions such as COUNT, SUM, AVG, MIN, and MAX summarize multiple rows.

Q22. COUNT(*) vs COUNT(column)?

Answer: COUNT(*) counts rows; COUNT(column) counts non-NULL values in that column.

Q23. What is a primary key in SQL?

Answer: A primary key uniquely identifies rows and enforces uniqueness and non-NULL semantics.

Q24. What is a constraint?

Answer: A constraint enforces rules such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK.

Q25. What is a CTE?

Answer: A Common Table Expression is a named temporary result set defined with WITH for a statement.

Q26. What is a window function?

Answer: A window function calculates across related rows without collapsing them into one row per group.

Q27. What is a view in SQL?

Answer: A view is a stored query presented as a virtual table.

Q28. UNION vs UNION ALL?

Answer: UNION removes duplicates; UNION ALL retains them and is usually faster.

Q29. What is NULL?

Answer: NULL represents missing or unknown data and must be handled with NULL-aware expressions.

Q30. What is an execution plan?

Answer: An execution plan shows how the database optimizer intends to execute a query.

Q31. What is COALESCE?

Answer: COALESCE returns the first non-NULL expression from its arguments.

Q32. What is CASE?

Answer: CASE provides conditional logic inside SQL expressions.

Q33. What is an alias?

Answer: An alias temporarily gives a column or table another name within a query.

Q34. What is query optimization?

Answer: Query optimization selects an efficient execution strategy for a SQL statement.

Q35. What is parameterized SQL?

Answer: Parameterized SQL separates query structure from input values, helping prevent injection and improve reuse.

4. Operating Systems

35 interview questions

Q1. What is an operating system?

Answer: An OS manages hardware and provides services and interfaces for applications.

Q2. What is a process?

Answer: A process is a program in execution with its own execution state and resources.

Q3. What is a thread?

Answer: A thread is a lightweight execution unit within a process.

Q4. Process vs thread?

Answer: Processes have separate address spaces; threads within a process typically share its memory and resources.

Q5. What is multitasking?

Answer: Multitasking allows multiple tasks to make progress through scheduling.

Q6. What is multiprocessing?

Answer: Multiprocessing uses multiple processing units or cores to execute tasks in parallel.

Q7. What is context switching?

Answer: Context switching saves one execution context and loads another so the CPU can switch tasks.

Q8. What is scheduling?

Answer: CPU scheduling selects which ready task should execute next.

Q9. What is FCFS?

Answer: First-Come, First-Served executes processes in arrival order.

Q10. What is SJF?

Answer: Shortest Job First selects the process with the smallest expected CPU burst.

Q11. What is Round Robin?

Answer: Round Robin gives each ready process a time quantum in cyclic order.

Q12. What is priority scheduling?

Answer: It selects processes according to assigned priority.

Q13. What is a race condition?

Answer: A race condition occurs when the result depends on the timing of concurrent operations.

Q14. What is a critical section?

Answer: A critical section is code that accesses shared data and needs controlled concurrent access.

Q15. What is a mutex?

Answer: A mutex provides mutual exclusion so only one thread owns a protected resource at a time.

Q16. What is a semaphore?

Answer: A semaphore is a synchronization primitive used to control access to shared resources.

Q17. What is deadlock?

Answer: Deadlock is a state where processes wait indefinitely for resources held by one another.

Q18. Four conditions for deadlock?

Answer: Mutual exclusion, hold and wait, no preemption, and circular wait.

Q19. Deadlock prevention vs avoidance?

Answer: Prevention breaks a necessary condition; avoidance allocates resources only when a safe state can be maintained.

Q20. What is virtual memory?

Answer: Virtual memory lets processes use an address space larger than physical RAM by using storage as backing.

Q21. What is paging?

Answer: Paging divides memory into fixed-size pages and frames.

Q22. What is segmentation?

Answer: Segmentation divides memory into logical variable-sized segments such as code or data.

Q23. What is a page fault?

Answer: A page fault occurs when a referenced page is not currently in physical memory.

Q24. What is thrashing?

Answer: Thrashing occurs when excessive paging leaves little CPU time for useful work.

Q25. What is fragmentation?

Answer: Fragmentation is wasted memory space; it can be internal or external.

Q26. What is a system call?

Answer: A system call is a controlled interface through which a program requests OS services.

Q27. Kernel vs shell?

Answer: The kernel manages core system resources; the shell provides a user interface to system commands.

Q28. What is booting?

Answer: Booting is the process of starting the system and loading the operating system.

Q29. What is IPC?

Answer: Inter-Process Communication enables processes to exchange data or synchronize.

Q30. What is a zombie process?

Answer: A zombie is a terminated child process whose exit status has not yet been collected by its parent.

Q31. What is starvation?

Answer: Starvation occurs when a process waits indefinitely because other tasks keep receiving resources or CPU time.

Q32. What is aging?

Answer: Aging gradually increases waiting processes' priority to reduce starvation.

Q33. What is a thread pool?

Answer: A thread pool maintains reusable worker threads to execute tasks efficiently.

Q34. What is demand paging?

Answer: Demand paging loads a page into memory only when it is needed.

Q35. What is a file system?

Answer: A file system organizes and manages files and directories on storage devices.

5. Computer Networks

35 interview questions

Q1. What is a computer network?

Answer: A network connects devices so they can communicate and share resources.

Q2. What is the OSI model?

Answer: OSI is a seven-layer conceptual model for network communication.

Q3. Name OSI layers.

Answer: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q4. What is TCP/IP model?

Answer: TCP/IP is the practical protocol architecture used by the Internet.

Q5. What is an IP address?

Answer: An IP address identifies a network interface or endpoint using an Internet Protocol address.

Q6. IPv4 vs IPv6?

Answer: IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses and provides a vastly larger address space.

Q7. What is a MAC address?

Answer: A MAC address is a link-layer hardware/interface identifier used on local networks.

Q8. What is a router?

Answer: A router forwards packets between different networks.

Q9. What is a switch?

Answer: A switch forwards frames within a local network, commonly using MAC addresses.

Q10. Hub vs switch?

Answer: A hub broadcasts traffic to ports; a switch selectively forwards frames.

Q11. What is DNS?

Answer: DNS translates domain names into IP addresses and other records.

Q12. What is DHCP?

Answer: DHCP automatically provides network configuration such as IP address, gateway, and DNS server.

Q13. What is HTTP?

Answer: HTTP is an application-layer protocol for transferring web resources.

Q14. HTTP vs HTTPS?

Answer: HTTPS is HTTP protected by TLS encryption and authentication.

Q15. What is TCP?

Answer: TCP is a connection-oriented transport protocol providing reliable, ordered byte-stream delivery.

Q16. What is UDP?

Answer: UDP is a connectionless transport protocol with low overhead and no built-in delivery guarantee.

Q17. TCP vs UDP?

Answer: TCP favors reliability and ordering; UDP favors lower overhead and latency.

Q18. What is a port?

Answer: A port identifies a logical service endpoint on a host.

Q19. What is a socket?

Answer: A socket is a communication endpoint identified by addressing information such as IP and port.

Q20. What is a subnet mask?

Answer: A subnet mask separates network and host portions of an IPv4 address.

Q21. What is subnetting?

Answer: Subnetting divides a network into smaller logical networks.

Q22. What is NAT?

Answer: Network Address Translation maps addresses between network address spaces, commonly private to public.

Q23. What is a firewall?

Answer: A firewall filters network traffic according to security rules.

Q24. What is latency?

Answer: Latency is the time taken for data or a packet to travel between endpoints.

Q25. What is bandwidth?

Answer: Bandwidth is the maximum data transfer capacity of a communication link.

Q26. What is packet loss?

Answer: Packet loss occurs when transmitted packets fail to reach the destination.

Q27. What is routing?

Answer: Routing determines paths for packets across interconnected networks.

Q28. What is ARP?

Answer: ARP maps an IPv4 address to a MAC address on a local network.

Q29. What is ICMP?

Answer: ICMP carries network control, diagnostic, and error messages.

Q30. What is the three-way handshake?

Answer: TCP establishes a connection using SYN, SYN-ACK, and ACK.

Q31. What is a proxy server?

Answer: A proxy acts as an intermediary between clients and destination servers.

Q32. What is CDN?

Answer: A Content Delivery Network distributes cached content through geographically distributed servers.

Q33. What is a URL?

Answer: A Uniform Resource Locator identifies the location and access method of a resource.

Q34. What is TLS?

Answer: Transport Layer Security provides encryption, integrity, and authentication for network communication.

Q35. What is ping?

Answer: Ping is a diagnostic utility commonly using ICMP echo messages to test reachability and round-trip time.

6. Data Structures & Algorithms

35 interview questions

Q1. What is a data structure?

Answer: A data structure organizes data for efficient access and modification.

Q2. What is an algorithm?

Answer: An algorithm is a finite sequence of steps used to solve a problem.

Q3. Array vs linked list?

Answer: Arrays provide indexed contiguous storage; linked lists use nodes connected by links.

Q4. What is a stack?

Answer: A stack follows LIFO: last in, first out.

Q5. What is a queue?

Answer: A queue follows FIFO: first in, first out.

Q6. What is a deque?

Answer: A deque allows insertion and deletion at both ends.

Q7. What is a linked list?

Answer: A linked list is a sequence of nodes connected through references or pointers.

Q8. Singly vs doubly linked list?

Answer: A singly list has a next link; a doubly list has next and previous links.

Q9. What is a tree?

Answer: A tree is a hierarchical, non-linear data structure.

Q10. What is a binary tree?

Answer: A binary tree allows each node to have at most two children.

Q11. What is a BST?

Answer: A Binary Search Tree maintains an ordering where left values are smaller and right values larger, subject to its definition.

Q12. What is a heap?

Answer: A heap is a complete-tree-based structure satisfying a heap-order property.

Q13. Min heap vs max heap?

Answer: A min heap keeps the smallest key at the root; a max heap keeps the largest.

Q14. What is a graph?

Answer: A graph consists of vertices and edges representing relationships.

Q15. BFS vs DFS?

Answer: BFS explores level by level using a queue; DFS explores deeply using recursion or a stack.

Q16. What is hashing?

Answer: Hashing maps keys to positions using a hash function for fast average lookup.

Q17. What is collision?

Answer: A collision occurs when different keys map to the same hash location.

Q18. What is binary search?

Answer: Binary search repeatedly halves a sorted search space; its typical time complexity is $O(\log n)$.

Q19. Linear vs binary search?

Answer: Linear search scans sequentially; binary search needs sorted data and is logarithmic.

Q20. What is sorting?

Answer: Sorting arranges data according to an ordering criterion.

Q21. Bubble sort complexity?

Answer: Typical time complexity is $O(n^2)$, with optimized best case $O(n)$.

Q22. Merge sort complexity?

Answer: Merge sort runs in $O(n \log n)$ time and uses additional memory for merging.

Q23. Quick sort complexity?

Answer: Average time is $O(n \log n)$; worst case is $O(n^2)$ with poor pivot choices.

Q24. What is recursion?

Answer: Recursion is when a function solves a problem by calling itself on smaller instances.

Q25. What is dynamic programming?

Answer: DP solves overlapping subproblems while storing results to avoid repeated work.

Q26. What is greedy algorithm?

Answer: A greedy algorithm repeatedly chooses the locally best available option.

Q27. What is Big-O?

Answer: Big-O describes an asymptotic upper bound on growth of time or space usage.

Q28. What is space complexity?

Answer: Space complexity measures extra memory required as input size grows.

Q29. What is a hash table?

Answer: A hash table stores key-value pairs using hashing for average constant-time lookup.

Q30. What is a trie?

Answer: A trie is a tree-like structure optimized for storing and searching strings by prefixes.

Q31. What is stable sorting?

Answer: A stable sort preserves the relative order of equal-key elements.

Q32. What is an adjacency matrix?

Answer: It represents graph edges using a matrix, often requiring $O(V^2)$ space.

Q33. What is an adjacency list?

Answer: It stores each vertex's neighboring vertices and is often space-efficient for sparse graphs.

Q34. What is memoization?

Answer: Memoization caches results of function calls to avoid repeated computation.

Q35. What is amortized analysis?

Answer: Amortized analysis studies average cost over a sequence of operations rather than each operation alone.

7. Software Engineering

35 interview questions

Q1. What is software engineering?

Answer: Software engineering applies systematic methods to develop, test, deploy, and maintain software.

Q2. What is SDLC?

Answer: SDLC is the Software Development Life Cycle covering activities from requirements through maintenance.

Q3. Common SDLC models?

Answer: Waterfall, Iterative, Spiral, V-Model, Agile, and DevOps-oriented approaches are common models.

Q4. What is Agile?

Answer: Agile emphasizes iterative delivery, customer feedback, collaboration, and adaptation.

Q5. What is Scrum?

Answer: Scrum is an Agile framework using roles, events, artifacts, and short iterations called sprints.

Q6. What is a sprint?

Answer: A sprint is a fixed period in Scrum during which a usable increment is developed.

Q7. What is a requirement?

Answer: A requirement describes a needed capability, constraint, or quality of a system.

Q8. Functional vs non-functional requirements?

Answer: Functional requirements describe what the system does; non-functional requirements describe qualities or constraints.

Q9. What is feasibility study?

Answer: It evaluates whether a proposed system is technically, economically, operationally, and otherwise viable.

Q10. What is SRS?

Answer: Software Requirements Specification documents system requirements and constraints.

Q11. What is testing?

Answer: Testing evaluates software to find defects and verify expected behavior.

Q12. Verification vs validation?

Answer: Verification asks whether the product is built correctly; validation asks whether the right product was built.

Q13. Unit testing?

Answer: Unit testing tests small components in isolation.

Q14. Integration testing?

Answer: Integration testing checks interactions among components or services.

Q15. System testing?

Answer: System testing evaluates the complete integrated system against requirements.

Q16. Acceptance testing?

Answer: Acceptance testing checks whether the system meets business or user acceptance criteria.

Q17. Black-box testing?

Answer: Black-box testing tests behavior without relying on internal implementation details.

Q18. White-box testing?

Answer: White-box testing uses knowledge of internal code or control flow.

Q19. Regression testing?

Answer: Regression testing checks that existing functionality still works after changes.

Q20. What is CI/CD?

Answer: CI/CD automates integration, testing, and delivery/deployment pipelines.

Q21. What is version control?

Answer: Version control tracks changes to source code and supports collaboration.

Q22. What is Git?

Answer: Git is a distributed version-control system.

Q23. What is code review?

Answer: Code review is systematic peer examination of code for correctness, quality, and maintainability.

Q24. What is technical debt?

Answer: Technical debt is the future cost created by choosing expedient solutions that require later improvement.

Q25. What is a bug?

Answer: A bug is a defect that causes software to behave incorrectly or unexpectedly.

Q26. What is debugging?

Answer: Debugging is the process of locating and fixing the cause of software defects.

Q27. What is maintenance?

Answer: Maintenance modifies software after delivery to fix, adapt, improve, or prevent issues.

Q28. What is risk management?

Answer: Risk management identifies, assesses, mitigates, and monitors project risks.

Q29. What is a design pattern?

Answer: A design pattern is a reusable solution template for a recurring software design problem.

Q30. Why documentation?

Answer: Documentation preserves knowledge about requirements, design, usage, operations, and maintenance.

Q31. What is an API contract?

Answer: It specifies how clients and services communicate, including inputs, outputs, errors, and rules.

Q32. What is a use case?

Answer: A use case describes how an actor interacts with a system to achieve a goal.

Q33. What is a prototype?

Answer: A prototype is an early model used to explore requirements, design, or usability.

Q34. What is scalability?

Answer: Scalability is the ability of a system to handle increased workload by adding or improving resources.

Q35. What is reliability?

Answer: Reliability is the ability of software to perform correctly and consistently over time.

8. Computer Organization

35 interview questions

Q1. What is computer organization?

Answer: It describes how computer components are structured and operate together.

Q2. What is CPU?

Answer: The CPU executes instructions and coordinates processing operations.

Q3. What is ALU?

Answer: The Arithmetic Logic Unit performs arithmetic and logical operations.

Q4. What is control unit?

Answer: The control unit directs instruction execution and coordinates CPU components.

Q5. What is a register?

Answer: A register is a small, fast storage location inside the CPU.

Q6. What is cache memory?

Answer: Cache is small, fast memory that stores frequently accessed data and instructions near the CPU.

Q7. RAM vs ROM?

Answer: RAM is generally writable working memory; ROM refers to non-volatile memory traditionally used for firmware, with modern variants.

Q8. What is virtual memory?

Answer: Virtual memory provides processes with a larger logical address space backed by physical memory and storage.

Q9. What is an instruction cycle?

Answer: It generally involves fetching, decoding, and executing an instruction, with additional steps depending on architecture.

Q10. What is a program counter?

Answer: The program counter stores the address of the next instruction to fetch.

Q11. What is an instruction register?

Answer: The instruction register holds the current instruction being decoded/executed.

Q12. What is pipelining?

Answer: Pipelining overlaps stages of multiple instructions to improve instruction throughput.

Q13. What is RISC?

Answer: Reduced Instruction Set Computer designs favor a smaller, simpler instruction set and regular execution.

Q14. What is CISC?

Answer: Complex Instruction Set Computer designs support a richer set of instructions, often with more complex operations.

Q15. What is clock speed?

Answer: Clock speed is the frequency at which a processor's clock cycles occur.

Q16. What is bus?

Answer: A bus is a communication pathway carrying data, addresses, or control signals.

Q17. Data bus vs address bus?

Answer: The data bus carries data; the address bus carries addresses used to identify locations.

Q18. What is interrupt?

Answer: An interrupt signals the CPU that an event needs attention.

Q19. What is DMA?

Answer: Direct Memory Access lets devices transfer data to/from memory with limited CPU involvement.

Q20. What is endianness?

Answer: Endianness defines the byte order used to represent multi-byte values in memory.

Q21. What is cache hit?

Answer: A cache hit occurs when requested data is found in the cache.

Q22. What is cache miss?

Answer: A cache miss occurs when requested data is not found in the cache.

Q23. What is locality of reference?

Answer: Programs tend to reuse recently accessed data and nearby addresses, enabling caching.

Q24. What is memory hierarchy?

Answer: Memory hierarchy arranges storage from fastest/smallest to slower/larger levels such as registers, cache, RAM, and storage.

Q25. What is throughput?

Answer: Throughput is the amount of useful work completed per unit time.

Q26. What is latency?

Answer: Latency is the delay before an operation's result becomes available.

Q27. What is an instruction set?

Answer: An instruction set is the set of machine instructions supported by a processor architecture.

Q28. What is addressing mode?

Answer: An addressing mode specifies how an instruction identifies its operands.

Q29. What is binary representation?

Answer: Binary representation encodes data using bits, typically 0 and 1.

Q30. What is two's complement?

Answer: Two's complement is a common representation for signed integers enabling convenient arithmetic.

Q31. What is superscalar execution?

Answer: A superscalar processor can issue multiple instructions per clock cycle when dependencies allow.

Q32. What is branch prediction?

Answer: Branch prediction guesses the outcome of control-flow branches to keep the pipeline busy.

Q33. What is cache associativity?

Answer: Associativity determines how memory blocks can be placed within cache sets.

Q34. What is an opcode?

Answer: An opcode specifies the operation that a machine instruction performs.

Q35. What is microarchitecture?

Answer: Microarchitecture is the internal implementation of a processor architecture.

9. Python / Java Fundamentals

35 interview questions

Q1. What is Python?

Answer: Python is a high-level, general-purpose programming language known for readability and a large ecosystem.

Q2. What is Java?

Answer: Java is a statically typed, class-based programming language designed around portability through the JVM.

Q3. Compiled vs interpreted?

Answer: Compilation translates source into another executable form; interpretation executes through a runtime. Modern languages often use a mix of techniques.

Q4. What is JVM?

Answer: The Java Virtual Machine executes Java bytecode and provides runtime services.

Q5. JDK vs JRE vs JVM?

Answer: JVM executes bytecode; JRE provides the runtime environment; JDK includes development tools plus the runtime.

Q6. Python list vs tuple?

Answer: Lists are mutable; tuples are immutable sequences.

Q7. What is a Python dictionary?

Answer: A dictionary stores key-value pairs with hash-based lookup.

Q8. What is a Python set?

Answer: A set stores unique elements and supports efficient membership operations.

Q9. What is mutability?

Answer: Mutable objects can change their contents after creation; immutable objects cannot.

Q10. What is exception handling?

Answer: Exception handling manages runtime errors using mechanisms such as try/except in Python and try/catch in Java.

Q11. What is inheritance in Java?

Answer: Java inheritance allows a class to extend another class and reuse or specialize behavior.

Q12. What is interface in Java?

Answer: A Java interface defines a contract that classes can implement.

Q13. What is garbage collection?

Answer: Garbage collection automatically reclaims memory that is no longer reachable.

Q14. == vs equals() in Java?

Answer: == compares primitive values or object references; equals() can compare object content when overridden appropriately.

Q15. What is method overloading in Java?

Answer: Methods can share a name when their parameter lists differ.

Q16. What is method overriding in Java?

Answer: A subclass supplies a compatible implementation of an inherited instance method.

Q17. What is a static member?

Answer: A static member belongs to the class rather than individual instances.

Q18. What is final in Java?

Answer: final can prevent reassignment of variables, overriding of methods, or inheritance of classes.

Q19. What is a lambda?

Answer: A lambda is a concise expression of a function-like behavior; Java supports lambdas for functional interfaces.

Q20. What is a Python lambda?

Answer: A Python lambda is a small anonymous function expression.

Q21. What is list comprehension?

Answer: A compact Python syntax for constructing lists from an iterable with optional filtering.

Q22. What is iterator?

Answer: An iterator produces elements one at a time, supporting sequential traversal.

Q23. What is generator in Python?

Answer: A generator lazily produces values, commonly using yield, saving memory for large sequences.

Q24. What is package?

Answer: A package organizes related modules/classes under a namespace.

Q25. What is module in Python?

Answer: A Python module is a file containing definitions and executable statements.

Q26. What is indentation in Python?

Answer: Indentation defines code blocks in Python.

Q27. What is garbage collection in Python?

Answer: Python uses reference counting plus cyclic garbage collection in its standard implementation.

Q28. What is pass-by-value?

Answer: Java passes argument values; for object references, the reference value itself is copied.

Q29. Why is Python dynamically typed?

Answer: Variable names are not declared with fixed types; objects carry types and names can refer to different types at different times.

Q30. What is type casting?

Answer: Type casting converts a value from one data type to another where supported.

Q31. What is a virtual environment in Python?

Answer: It isolates project dependencies and Python packages from other projects.

Q32. What is a decorator in Python?

Answer: A decorator wraps a function or class to extend or modify behavior.

Q33. What is an abstract class in Java?

Answer: A class declared abstract that may contain abstract and concrete methods and cannot be instantiated directly.

Q34. What is an exception vs error?

Answer: An exception is a condition a program may handle; an error often denotes a serious or system-level problem, depending on language terminology.

Q35. What is an enum?

Answer: An enum defines a fixed set of named constants.

10. Web Technologies

35 interview questions

Q1. What is HTML?

Answer: HTML structures content on web pages using elements and attributes.

Q2. What is CSS?

Answer: CSS controls presentation, layout, and visual styling of HTML documents.

Q3. What is JavaScript?

Answer: JavaScript is a programming language widely used to add behavior and interactivity to web applications.

Q4. HTML vs CSS?

Answer: HTML defines structure/content; CSS defines presentation/layout.

Q5. What is semantic HTML?

Answer: Semantic HTML uses elements that describe the meaning of content, improving accessibility and structure.

Q6. What is DOM?

Answer: The Document Object Model represents an HTML document as a tree of objects that scripts can manipulate.

Q7. What is responsive design?

Answer: Responsive design adapts layout and content to different screen sizes and devices.

Q8. What is a CSS selector?

Answer: A selector targets HTML elements for styling.

Q9. What is the CSS box model?

Answer: The box model consists of content, padding, border, and margin.

Q10. Flexbox vs Grid?

Answer: Flexbox is mainly one-dimensional; Grid is designed for two-dimensional row-and-column layouts.

Q11. What is an API?

Answer: An API is an interface that allows software components to communicate using defined rules.

Q12. What is REST?

Answer: REST is an architectural style commonly used for web APIs using resources and standard HTTP methods.

Q13. GET vs POST?

Answer: GET retrieves data; POST commonly submits data for processing or creation.

Q14. PUT vs PATCH?

Answer: PUT commonly replaces a resource representation; PATCH applies partial modifications.

Q15. What is JSON?

Answer: JSON is a lightweight text format commonly used to exchange structured data.

Q16. What is HTTP status code 200?

Answer: 200 indicates a successful request.

Q17. What is 404?

Answer: 404 indicates that the requested resource could not be found.

Q18. What is 500?

Answer: 500 indicates a generic server-side error.

Q19. What is authentication?

Answer: Authentication verifies who a user or system is.

Q20. What is authorization?

Answer: Authorization determines what an authenticated user or system is allowed to do.

Q21. Cookies vs localStorage?

Answer: Cookies are sent with applicable HTTP requests; localStorage stores data in the browser and is not automatically sent to servers.

Q22. What is CORS?

Answer: Cross-Origin Resource Sharing controls whether browsers allow requests from one origin to another.

Q23. What is frontend?

Answer: Frontend is the client-side interface users interact with in a web application.

Q24. What is backend?

Answer: Backend handles server-side logic, data access, authentication, and APIs.

Q25. What is a database connection?

Answer: It is a communication channel between an application and a database system.

Q26. What is form validation?

Answer: Form validation checks user input for required format, range, and business rules.

Q27. What is XSS?

Answer: Cross-Site Scripting is an attack where untrusted script content executes in a user's browser.

Q28. What is SQL injection?

Answer: SQL injection occurs when untrusted input alters a database query; parameterized queries help prevent it.

Q29. What is HTTPS?

Answer: HTTPS protects HTTP communication using TLS encryption and server authentication.

Q30. What is caching in web apps?

Answer: Caching stores reusable data or responses closer to users or applications to reduce latency and server work.

Q31. What is event bubbling?

Answer: An event can propagate from a target element upward through its ancestor elements.

Q32. What is event delegation?

Answer: Event delegation handles events on a parent instead of attaching separate handlers to many children.

Q33. What is semantic accessibility?

Answer: Using meaningful structure and labels helps assistive technologies interpret web content.

Q34. What is lazy loading?

Answer: Lazy loading delays loading non-critical resources until they are needed.

Q35. What is a session?

Answer: A session represents server-side or application-level state associated with a user's interaction period.